

FINITE-DEPTH COLLATZ CERTIFICATION THROUGH SHARP FIRST-CROSSING ENVELOPES

ABSTRACT. We study the gap between coefficient contraction and actual descent in the shortcut Collatz map. The first-contraction constraint determines a unique rightmost feasible parity word, whose affine offset is maximal. This sharp envelope reduces certification at depth K to an explicit finite set of potential exceptions, without enumerating 2^K residue classes. At $K = 1024$, exact arithmetic gives the exception bound 72058. Independent verification of this finite range proves that every integer $n > 1$ with coefficient stopping time at most 1024 has equal actual stopping time. An independent first-passage recurrence counts the unclassified residue mass, approximately $1.1537401 \times 10^{-19}$ at this depth. We also derive the asymptotic normalized envelope $1/(6 \log 2)$ and show why neither a fixed depth nor a fixed envelope scan limit proves universal descent. This is a self-contained computational study of established parity-vector methods; no priority claim or resolution of the general conjecture is made.

1. QUESTION, CONTEXT, AND PRINCIPAL CONCLUSION

Let

$$(1) \quad T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}, \end{cases}$$

on the positive integers. Write $q_j(n)$ for the number of odd terms among the first j iterates, including n and excluding $T^j(n)$, and define

$$C_j(n) = \frac{3^{q_j(n)}}{2^j}, \quad \tau(n) = \min\{j \geq 1 : C_j(n) < 1\}, \quad \sigma(n) = \min\{j \geq 1 : T^j(n) < n\}.$$

An empty minimum is ∞ . All statements about equality of these stopping times below exclude $n = 1$, for which $\tau(1) = 2$ and $\sigma(1) = \infty$.

The conjecture that $\sigma(n) = \tau(n)$ for all $n \geq 2$ is usually called the coefficient stopping time conjecture, following Terras [3]. Rozier and Terracol [1] discuss its relation to finite trajectories with coefficient below one but endpoint at least the starting value. Their example 7, 11, 17, 26, 13, 20, 10, 5, 8 illustrates why contraction at an arbitrary later time is insufficient: the coefficient at step 8 is $243/256 < 1$, yet the endpoint is 8. This does not violate the conjecture, because descent already occurred at step 7.

The present question is computational and conditional: can the equality be certified at a large *first-contraction depth*, uniformly over all starting integers, without constructing the entire parity tree? The principal finite conclusion is the following.

Theorem 1.1 (Exact finite-depth certificate). *For every positive integer $n > 1$,*

$$\tau(n) \leq 1024 \implies \sigma(n) = \tau(n).$$

There is no size restriction on n . The implication makes no assertion when $\tau(n) > 1024$ or $\tau(n) = \infty$.

The proof combines an elementary sharp envelope, a finite reduction, and the accompanying exact certificate checks. A complete derivation follows; the numerical part is isolated in Section 5. Parity classes and offset comparisons are established tools, and the

Date: October 9, 2026.

2020 Mathematics Subject Classification. Primary 11B83; Secondary 37P99.

Key words and phrases. Collatz iteration, coefficient stopping time, mechanical word, finite certificate.

literature reports much larger computational investigations than the small scan used here. The purpose is a transparent, independently replayable reduction, not a new verification record or a claim that the general stopping-time conjecture has become tractable.

2. AFFINE IDENTITIES AND FIRST-CROSSING CONSTRAINTS

For a binary word $w = (\varepsilon_0, \dots, \varepsilon_{k-1})$, let $a_j = \sum_{i < j} \varepsilon_i$. On the integers realizing this parity word,

$$(2) \quad T^j(n) = \frac{3^{a_j}n + B_j}{2^j}, \quad B_0 = 0,$$

where

$$(3) \quad B_{j+1} = \begin{cases} B_j, & \varepsilon_j = 0, \\ 3B_j + 2^j, & \varepsilon_j = 1. \end{cases}$$

Each word is realized by exactly one residue class modulo 2^k . To see this, suppose a word of length j has residue r . Its two lifts r and $r + 2^j$ have j th iterates differing by the odd integer 3^{a_j} , so they realize opposite next parities. Induction proves both the residue assertion and (2). All offsets are nonnegative. Hence actual descent requires coefficient contraction, and

$$(4) \quad \sigma(n) \geq \tau(n).$$

Set $\alpha = \log_3 2$ and define the exact barrier

$$(5) \quad h_j = \min\{a \geq 0 : 3^a \geq 2^j\}, \quad h_0 = 0.$$

For $j > 0$, $h_j = \lceil \alpha j \rceil$; equality of positive powers of 2 and 3 is impossible. Since $0 < \alpha < 1$, successive barrier increments are either zero or one. In implementation, (5) is evaluated by integer comparisons, never by rounding logarithms.

Lemma 2.1. *If a word first contracts at depth k , its last bit is zero and its total odd count is $q = h_{k-1}$. Such a word exists exactly when*

$$(6) \quad 2^{k-1} \leq 3^{h_{k-1}} < 2^k.$$

All its proper prefix counts obey $a_j \geq h_j$ for $j < k$.

Proof. An odd step multiplies the coefficient by $3/2$, and therefore cannot be the first contraction. At the preceding depth the coefficient is at least one, so $2^{k-1} \leq 3^q < 2^k$. This interval contains at most one power of 3 and, if it contains one, its exponent is h_{k-1} . Noncontraction of each earlier prefix gives the barrier condition. Conversely, the word whose prefix counts are h_j for $j < k$, followed by a zero bit, realizes first contraction whenever (6) holds. All these counts are achievable because barrier increments are binary. $\square$

3. A SHARP CONSTRAINED OFFSET ENVELOPE

For every feasible k , let w_k^* have proper prefix counts exactly h_j and final bit zero. Denote its offset by M_k . This construction postpones each odd step as far as the prefix barriers permit.

Theorem 3.1 (Sharp first-crossing envelope). *Among all words first contracting at a feasible depth k , w_k^* is the unique word with maximal offset. If $q = h_{k-1}$ and the odd positions of a word are $i_1 < \dots < i_q$, then*

$$(7) \quad B_k = \sum_{\ell=1}^q 3^{q-\ell} 2^{i_\ell}, \quad i_\ell \leq i_\ell^* := \lfloor (\ell-1) \log_2 3 \rfloor.$$

Thus $B_k \leq M_k$, with equality only for w_k^ . For $q = 0$, the sole word is the even one-step word and $M_1 = 0$.*

Proof. Each odd step inserts 2^{i_ℓ} in (3); each later odd step multiplies that contribution by 3. Summing gives the first identity. The barrier first reaches ℓ at

$$j_\ell = \min\{j : h_j \geq \ell\} = \lfloor (\ell - 1) \log_2 3 \rfloor + 1.$$

Every feasible word must already have ℓ odd positions among its first j_ℓ bits. Therefore $i_\ell \leq j_\ell - 1 = i_\ell^*$. The proposed word attains all these upper bounds simultaneously. Every summand in (7) strictly increases with its position, proving maximality and uniqueness. This argument does not require a sequence of swaps to remain feasible at every intermediate word. $\square$

A complete small example. At $k = 8$, the barrier values $(h_0, \dots, h_8)$ are $(0, 1, 2, 2, 3, 4, 4, 5, 6)$. The extremal word is 11011010, with odd positions 0, 1, 3, 4, 6, so

$$M_8 = 3^4 + 3^3 2 + 3^2 2^3 + 3 2^4 + 2^6 = 319.$$

The word 11111000 is also a first-crossing word, but its earlier odd positions give offset 211. Both have $q = 5$ and coefficient $243/256$ at depth 8. Figure 1 exhibits the prefix constraint responsible for their different offsets.

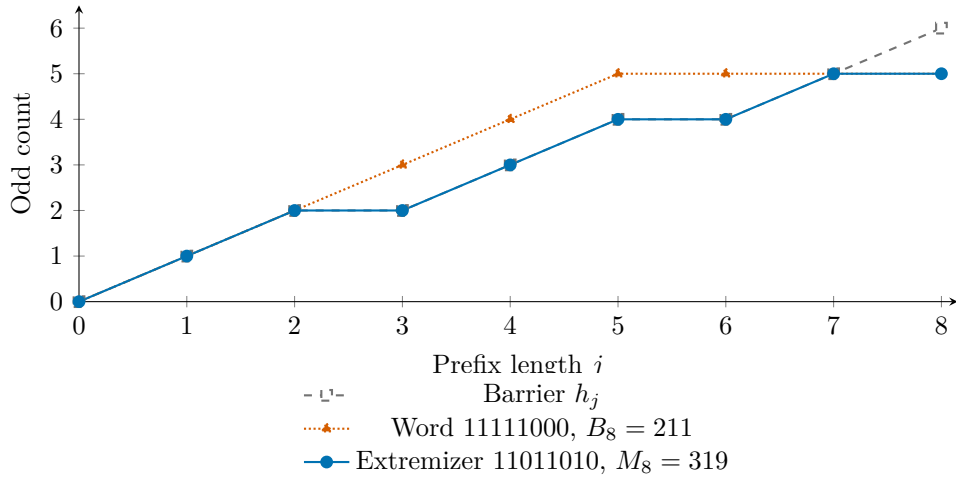


FIGURE 1. First crossing at depth 8. Counts must stay at or above the barrier at every proper prefix and fall below it at the last step. Lines connect integer-depth observations only. Delaying odd positions within these constraints maximizes the affine offset; an unconstrained late-odd word need not obey the barriers.

An exhaustive check of the 256 residues modulo 256 gives seven words with first crossing at depth 8 and confirms that the largest offset is 319. The two illustrated words correspond respectively to residues 123 and 95. In particular, a positive integer realizing the extremal word is at least 123, whereas its failure bound is $\lfloor 319/(256 - 243) \rfloor = 24$. Thus that class contains no failure. This example also shows why the uniform envelope is a sufficient bound: it discards the residue restriction, which may sharpen individual cases.

The unconstrained maximum for fixed length k and odd count q instead puts all odd bits at the end. It equals

$$(8) \quad U_{k,q} = 2^{k-q}(3^q - 2^q).$$

Indeed, the largest possible odd positions are $k - q, \dots, k - 1$; substitution in (7) gives (8). That word normally contracts long before its last step, so the bound throws away the crucial first-crossing information. At $k = 485$, $q = 306$, its ratio to M_k is approximately 1.03971×10^{52} . This is a comparison of mathematical bounds, not a measured runtime speedup against all existing Collatz algorithms.

Theorem 3.2 (Finite reduction). *For an integer $K \geq 1$, define*

$$(9) \quad H_K = \max_{\substack{k \leq K \\ k \text{ feasible}}} \left\lfloor \frac{M_k}{2^k - 3^{h_{k-1}}} \right\rfloor.$$

If every integer $2 \leq n \leq H_K$ with $\tau(n) \leq K$ satisfies $T^{\tau(n)}(n) < n$, then every $n > 1$ with $\tau(n) \leq K$ satisfies $\sigma(n) = \tau(n)$.

Proof. Let $\tau(n) = k \leq K$ and put $A = 3^{h_{k-1}}$, $D = 2^k$. By (2) and Theorem 3.1,

$$T^k(n) \geq n \implies (D - A)n \leq B_k \leq M_k \implies n \leq H_K.$$

The hypothesized finite check excludes such a failure for $n > 1$. Therefore descent occurs at time k . Equation (4) rules out an earlier descent and yields equality of the stopping times. The weak inequality at the failure boundary is essential: the floor in (9) must include equality, not discard it as a strict comparison. $\square$

4. COMPRESSED COUNTING AND INDEPENDENT CHECKS

For counting purposes only, write $s_{j,a}$ for the number of length- j words having a odd bits and no contracted prefix. Initialize $s_{0,0} = 1$. Each state contributes to odd counts a and $a + 1$ at the next depth; retain the contribution precisely if $3^a \geq 2^{j+1}$ or $3^{a+1} \geq 2^{j+1}$, respectively. Let F_j count words whose first contraction occurs at j , and $R_K = \sum_a s_{K,a}$. The disjoint first-passage decomposition gives

$$(10) \quad R_K = 2^K - \sum_{j=1}^K F_j 2^{K-j}, \quad \rho_K = \frac{R_K}{2^K}.$$

By the parity-residue bijection, ρ_K is the natural density of $\{n : \tau(n) > K\}$. It is not a sampled frequency or a probability of divergence. After finite-depth equality is certified, it is also the density of $\{n : \sigma(n) > K\}$; deleting the exceptional point 1 does not change a natural density.

The production code computes barriers, extremal offsets, and forward counts using integer arithmetic. There are $O(K^2)$ counting-state updates, instead of enumerating all 2^K words. This counts arithmetic operations; integers have increasing bit length, so it is not an $O(K^2)$ bit-complexity claim. Explicit certificate rows retain the large integers needed to replay the thresholds.

The checker uses two different recurrences. For extrema, it keeps the maximum offset in each surviving state (j, a) , applies the affine update to both next bits, and records maxima at first crossings. This max-plus calculation does not use the rightmost-word formula. For counts, it uses binomial renewal instead of the forward state recursion. At a feasible depth k , with $q_k = h_{k-1}$,

$$(11) \quad F_k = \binom{k-1}{q_k} - \sum_{\substack{j < k \\ j \text{ feasible}}} F_j \binom{k-1-j}{q_k - q_j},$$

where a binomial coefficient outside its usual range is zero. The first term counts all prefixes of length $k - 1$ having q_k odd bits. The sum removes those with an earlier first crossing; their remaining bits are arbitrary. Appending a zero produces the crossing at k . For an infeasible k , $F_k = 0$.

Finite trajectories are checked separately: iterate each integer through its first coefficient contraction and compare the actual endpoint with the initial value. Reaching the cap without coefficient contraction is reported outside the theorem, not as a counterexample or a success. The checker also verifies complete depth and domain coverage and rejects six damaged certificates: wrong offset, wrong threshold, missing crossing depth, wrong density, shortened scan domain, and altered scan histogram. The verifier uses explicit

exceptions rather than assertion-dependent acceptance. An author-independent review additionally rederived the extremal bound, checked all first-crossing words through depth 18, and recomputed the depth-1024 extrema and finite trajectories.

5. EXACT RESULTS

TABLE 1. Finite-depth envelopes and exact-count residue densities. The densities are displayed approximately; the certificate retains exact integer numerators and denominators.

K	H_K	Depth attaining H_K	ρ_K
16	24	8	3.2257080×10^{-2}
32	108	27	9.6269611×10^{-3}
64	281	46	1.4930654×10^{-3}
128	867	65	6.9777999×10^{-5}
256	4862	233	3.2760221×10^{-7}
512	72058	485	$1.6152967 \times 10^{-11}$
1024	72058	485	$1.1537401 \times 10^{-19}$

Figure 2 separates two effects of increasing depth: the remaining residue density falls, while the sufficient scan bound jumps when a new extremal depth is encountered. Density and scan size measure different objects and are not combined into a single accuracy score.

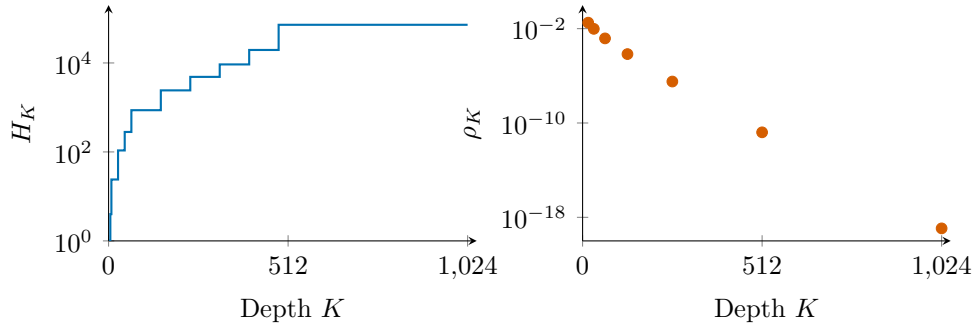


FIGURE 2. Two exact-count summaries from the depth-1024 certificate, shown on separate logarithmic scales. Left: the sufficient exception bound is a step function (the zero value at $K = 1$ is omitted on the log scale). Right: unclassified natural density at the seven tabulated checkpoints; no interpolation is implied. The small final density is positive and does not establish universal descent.

At depth 1024 there are 647 feasible first-crossing lengths. The largest exception floor occurs at $(k, q) = (485, 306)$ and is 72058. For that pair, the unconstrained offset bound (8) would give a scan limit of approximately 7.49195×10^{56} . The reduction to 72058 is the practical role of the prefix constraint. The finite scan examines all 72057 integers from 2 through 72058. There are no endpoint failures and no capped trajectories. The largest coefficient stopping time in this scan is 135, attained only at 35655; this is a statistic of the finite scan, not an upper bound for all integers in Theorem 1.1.

Proof of Theorem 1.1. The exact certificate gives $H_{1024} = 72058$. The independent max-plus recurrence confirms every extremal offset and every threshold at the 647 feasible depths. Direct finite replay checks the hypothesis of Theorem 3.2 on every required initial

integer. That theorem gives the asserted implication uniformly over all $n > 1$. The certificate object, checking program, and completed run are part of the finite computational evidence, not a replacement for the preceding mathematical reduction. $\square$

The generating calculation took about 0.065 seconds, including the finite scan and 387646 forward state updates. The independently coded checker took about 0.639 seconds, including max-plus extrema, binomial renewal, finite replay, and negative controls. These are observations on the recorded Python 3.12 environment, not universal benchmark timings. They exclude derivation, literature reading, review, and manuscript work. Fast arithmetic does not by itself indicate a deep mathematical result.

The earlier depth-16 explicit tree is retained as a separate baseline. Its 791 leaf words and 2114 unresolved residues agree exactly with the compressed count. The new result changes the scope from an explicit small tree and an unrelated finite scan to a uniform first-depth theorem; it does not merely enlarge the initial integer range.

6. ENVELOPE ASYMPTOTICS AND LIMITS OF CONTINUATION

The extremizer also describes how the sufficient scan bound behaves. Set $\beta = \log_2 3$, and for $q \geq 1$ let $k_q = \lfloor q\beta \rfloor + 1$. These are exactly the feasible crossing lengths other than $k = 1$. Put $\delta_q = k_q - q\beta \in (0, 1)$.

Proposition 6.1. *For $q \geq 1$,*

$$(12) \quad \frac{M_{k_q}}{3^q} = \frac{1}{3} \sum_{\ell=0}^{q-1} 2^{-\{\ell\beta\}}, \quad \frac{M_{k_q}}{q3^q} \longrightarrow \frac{1}{6 \log 2}.$$

In particular,

$$(13) \quad \left\lfloor \frac{M_{k_q}}{2^{k_q} - 3^q} \right\rfloor > \frac{q}{6} - 1.$$

Thus the sufficient scan limit H_K is unbounded as $K \rightarrow \infty$. This does not assert that any genuine exception exists.

Proof. Divide (7) at the maximizing positions by 3^q . Since $2^\beta = 3$, its ℓ th summand becomes $2^{-\{(\ell-1)\beta\}}/3$, proving the identity. The number β is irrational: a rational value would give equality of nontrivial integer powers of 2 and 3. For any nonzero integer m , the average of $\exp(2\pi i m \ell \beta)$ tends to zero by the finite geometric-series formula. Weyl's criterion therefore gives uniform distribution of $\{\ell\beta\}$; see [2], Proposition 1 and Exercise 5. Applying it to the bounded Riemann-integrable function 2^{-x} on $[0, 1)$ gives

$$\frac{1}{3q} \sum_{\ell < q} 2^{-\{\ell\beta\}} \longrightarrow \frac{1}{3} \int_0^1 2^{-x} dx = \frac{1}{6 \log 2}.$$

The single endpoint discontinuity on the circle is harmless: continuous upper and lower approximations can differ only on an interval of arbitrarily small length, whose asymptotic frequency equals its length. Every summand in the normalized sum is greater than $1/6$, and $2^{\delta_q} - 1 < 1$. Consequently

$$\frac{M_{k_q}}{2^{k_q} - 3^q} = \frac{M_{k_q}/3^q}{2^{\delta_q} - 1} > \frac{q}{6}.$$

Taking the floor proves (13) and unboundedness. $\square$

For illustration, $M_{k_q}/(q3^q)$ equals approximately 0.249815 at $q = 17$, 0.240844 at $q = 306$, and 0.240668 at $q = 646$; the limit is approximately 0.240449. These finite values illustrate the proved limit, but neither establish a finite error rate nor prove monotonic convergence. The denominator $2^{\delta_q} - 1$ explains large jumps of the envelope when a power of 3 is close to a power of 2 from below.

Proposition 6.2 (No fixed-horizon coverage). *For every $K \geq 1$, there is an $n > 1$ with $\tau(n) > K$ and $\sigma(n) > K$.*

Proof. For $K \geq 2$, take $n = 2^K - 1$. Direct induction gives

$$T^j(n) = 3^j 2^{K-j} - 1 \quad (0 \leq j \leq K).$$

The terms before the last are odd, and each term with $j \geq 1$ is larger than n . The coefficient is $(3/2)^j > 1$. For $K = 1$, take $n = 3$. $\square$

Thus the very small density in Table 1 cannot be rounded to zero or read as universal convergence. Nor can the finite theorem be iterated without an argument controlling the unclassified starting values. Proposition 6.1 identifies another limitation: this particular uniform envelope cannot certify all crossing depths by one fixed finite scan. It does not prove that other methods fail, and it does not show that the coefficient stopping time conjecture is false.

7. RESEARCH STATUS AND REPRODUCIBILITY

The study has three different evidence types: elementary proofs of the envelope and reduction, exact finite checks for the depth-1024 instance, and a qualitative asymptotic deduction from classical equidistribution. The recorded decimals are presentation values; all certificate decisions use integers. No random sampling, learned model, or fitted threshold is used. No proof assistant verifies the Python runtime or the entire paper.

From the revision directory, the standard-library entry points are

```
python code/certify.py
python code/verify.py
```

The certificate records every feasible depth, offset, exact threshold, density fraction, and finite scan summary. Separate review artifacts contain the author-independent calculations. Version hashes and the actual environment accompany the manuscript; generating a new run should be done in a copy if the sealed research record is to remain unchanged.

The main unresolved issue is structural. The all-integer conjecture needs control beyond every fixed first-crossing depth, or a genuinely different argument for the residual cases. Enlarging 1024 without such an argument would produce another finite certificate, not resolve that issue. The present continuation therefore stops after the validated reduction and its limitations. Originality of the constrained envelope as a standalone result has not been established; the local study is presented as a verified synthesis and computational note rather than a new general Collatz theorem.

REFERENCES

- [1] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, Discrete Math. **349** (2026), Article 115167. arXiv:2502.00948v5.
- [2] T. Tao, *254B, Notes 1: Equidistribution of polynomial sequences in tori*, lecture notes, March 28, 2010, author's online notes.
- [3] R. Terras, *A stopping time problem on the positive integers*, Acta Arith. **30** (1976), 241–252. doi:10.4064/aa-30-3-241-252.